Option-Implied Information, Machine Learning and Forecast Combinations in Realized Volatility Forecasting

Course title: Economic Forecasting (P)
Subject area: Volatility Forecasting

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Hand-in date: 15 May 2026

Specification of whether the assignment can be made public:
The assignment can be made public

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1. Introduction

Forecasting equity market volatility is central to risk management, portfolio allocation and derivative pricing. Realized volatility constructed from high-frequency data has become a standard measure of ex post return variation, while the HAR-RV model remains a widely used benchmark because it captures volatility persistence in a simple and interpretable way (Andersen et al., 2001; Andersen et al., 2003; Corsi, 2009). However, historical volatility dynamics may not contain all relevant information about future volatility, especially when market expectations and risk premia change rapidly.

Option prices provide a natural forward-looking source of information because they reflect expectations under the risk-neutral measure and compensation for volatility and tail risk. Implied volatility has been shown to predict future realized volatility beyond historical measures, and richer information from the implied volatility surface may further improve forecasts (Christensen and Prabhala, 1998; Busch et al., 2011; Kambouroudis et al., 2021; Michael et al., 2025). Macro-financial predictors may also be relevant at the monthly horizon because they capture business-cycle conditions, financial stress and aggregate uncertainty (Engle et al., 2013; Ludvigson and Ng, 2007; Nonejad, 2017).

This paper studies whether option-implied and macro-financial variables improve forecasts of next-month S&P 500 realized variance beyond historical volatility dynamics. The empirical framework compares a HAR benchmark with machine learning models estimated on option-implied variables, macro-financial predictors and their combination. It also considers equal-weighted and stacked forecast combinations, allowing forecasts from different models and information sets to be pooled rather than selecting a single specification ex ante.

Finally, the paper evaluates whether statistical forecast gains translate into economic value. Forecasts are used to construct an ex-ante variance-risk-premium signal, defined as implied variance minus forecasted realized variance, which scales a stylised short-volatility exposure. This links forecast accuracy to an investor timing problem while keeping the economic exercise transparent and comparable across models.

The results indicate that option-implied variables provide the clearest incremental forecasting value. HAR + Option specifications perform strongly under both R^2_{OOS} and QLIKE, while forecast combinations are also competitive. Macro-financial predictors are theoretically relevant, but their incremental value is less stable once option-implied information is included. Overall, monthly volatility forecasts benefit most from combining historical realized-variance dynamics with forward-looking option-market information.

2. Realized Volatility

Realized volatility provides a non-parametric measure of ex post return variation constructed from high-frequency intraday data. Rather than modelling volatility as a latent conditional variance, realized measures use the information contained in intraday returns to approximate the quadratic variation of the underlying price process. Following Andersen et al. (2001, 2003), realized variance is typically computed as the sum of squared intraday returns over a given trading period and has become a standard empirical proxy for latent volatility in financial econometrics.

The use of high-frequency data, however, involves a trade-off. Sampling at very high frequencies increases the amount of information used in the estimator, but it also makes the measure more sensitive to market microstructure noise, including bid-ask bounce and non-synchronous trading. For this reason, empirical studies often rely on sparse intraday sampling schemes, with five-minute returns representing a common compromise between efficiency and robustness. Liu et al. (2015) show that five-minute realized volatility is difficult to outperform across a broad set of assets, which supports its continued use as a practical benchmark measure.

2.1. Time-Series Dynamics of Volatility

A central feature of financial volatility is its strong persistence. Periods of elevated volatility tend to be followed by further periods of high volatility, while tranquil periods are also persistent. Empirical studies using realized measures document slowly decaying autocorrelations and long-memory-like behaviour, making autoregressive structures a natural starting point for volatility forecasting (Andersen et al., 2001; Andersen et al., 2003).

The heterogeneous autoregressive model of realized volatility, HAR-RV, proposed by Corsi (2009), captures this persistence through realized-volatility components measured over different horizons. In its standard form, the model combines short-, medium- and long-term components, reflecting the idea that market participants operate at heterogeneous investment horizons. Despite its simple linear structure, HAR-RV remains a strong benchmark in realized-volatility forecasting. Clements and Preve (2021) show that practical choices such as transformations, estimation methods and forecast combinations materially affect HAR performance, while Branco et al. (2024) find that nonlinear machine learning methods do not systematically dominate linear models across global stock indices. This makes HAR-RV an appropriate benchmark for evaluating richer information sets and more flexible forecasting methods.

2.2. Option-Implied Information and Volatility Forecasting

Option prices provide a forward-looking source of information about future market uncertainty. While realized volatility is constructed from past returns under the physical measure, option-implied measures are extracted from derivative prices and therefore reflect expectations under the risk-neutral measure as well as compensation for volatility and tail risk. This makes option-implied variables natural predictors of future realized volatility (Figlewski, 1997; Christensen and Prabhala, 1998; Jackwerth, 2000; Busch et al., 2011).

A large empirical literature shows that implied volatility contains information beyond historical volatility. Christensen and Prabhala (1998) find that implied volatility outperforms historical measures in forecasting future volatility, while Busch et al. (2011) document incremental predictive power across asset classes. More recent contributions demonstrate that HAR-type models can be improved by incorporating option-implied measures, including richer representations of the implied volatility surface (Kambouroudis et al., 2021; Michael et al., 2025).

The predictive content of option data extends beyond the level of implied volatility. At-the-money implied volatility captures overall expected volatility but embeds a variance risk premium due to risk-neutral pricing (Carr and Wu, 2009; Bollerslev et al., 2009; Bekaert and Hoerova, 2014). Measures of skew capture asymmetry in the risk-neutral distribution and are closely related to crash risk and demand for downside protection (Aït-Sahalia and Lo, 2000; Bakshi et al., 2003; Gârleanu et al., 2009; Bollerslev and Todorov, 2011).

Higher-order features of the implied volatility surface, such as curvature, provide information about tail risk and the pricing of extreme outcomes. Zhang and Xiang (2008) link the shape of the volatility smile to higher moments of the risk-neutral distribution, while Bakshi et al. (2003) show how option prices can be used to extract skewness and kurtosis. Dispersion measures capture heterogeneity in implied volatility across contracts, and recent work highlights that the construction of the implied volatility surface itself can affect forecasting performance (Ulrich et al., 2023; Michael et al., 2025).

Finally, the variance risk premium measures the difference between risk-neutral expected variance and expected realized variance and is commonly interpreted as compensation for volatility risk (Carr and Wu, 2009; Drechsler and Yaron, 2011; Bekaert and Hoerova, 2014). Taken together, these variables capture expected volatility, downside risk, tail risk and volatility risk compensation, providing a rich information set for forecasting realized volatility (Byun and Kim, 2013; Yfanti and Karanasos, 2022).

2.3. Macro-Financial Predictors and Volatility

Macro-financial variables may contain predictive information about future equity market volatility because volatility is linked to business-cycle conditions, financial stress and aggregate uncertainty. Changes in expected cash flows, discount rates and risk premia affect asset prices and their variability, making macro-financial conditions relevant for volatility forecasting (Campbell, 1996; Cochrane, 2011). Empirical evidence also shows that macroeconomic fundamentals and factors extracted from large datasets can help explain risk, returns and volatility, particularly at lower frequencies (Ludvigson and Ng, 2007; Engle et al., 2013; Nonejad, 2017; Gupta et al., 2023; Song et al., 2023).

Because macro-financial datasets are high-dimensional and strongly correlated, factor extraction and regularisation methods are commonly used to extract predictive information (Stock and Watson, 2002; McCracken and Ng, 2016; Zou and Hastie, 2005). An important empirical question is therefore whether macro-financial variables add information beyond option-implied measures, which may already incorporate market expectations about macroeconomic risk and uncertainty (Yfanti and Karanasos, 2022).

3. Data

3.1. Realized Volatility Construction

Realized-volatility data are obtained from Dacheng Xiu's Risk Lab (Xiu, n.d.), which provides realized measures for the S&P 500/SPX constructed from high-frequency intraday returns. Following Andersen et al. (2001), daily realized variance is computed as the sum of squared five-minute intraday returns,

$$RV_d = \sum_{i=1}^{M_d} r_{d,i}^2 ,$$

where $r_{d,i}$ denotes the i-th intraday return on day d, and M_d is the number of intraday intervals. Throughout the empirical analysis, RV denotes realized variance. The term realized volatility is used more broadly when referring to the realized-volatility literature and to realized measures of return variation.

The monthly realized variance target is constructed by aggregating daily realized variance within each month,

$$RV_t = \sum_{d \in t} R\,V_d.$$

This measure captures total realized return variation during month t and serves as the forecasting target. The models are estimated using the log transformation,

$$y_t = \log\left(RV_t\right),$$

which reduces the strong right skewness of realized variance and ensures that fitted forecasts can be retransformed to positive variance forecasts before evaluation.

3.2. Option-Implied Variable Construction

Option-implied variables are constructed from 30-day SPX implied volatilities across delta-based slices of the option surface. At-the-money 50-delta implied volatility is used to measure the general level of market-implied volatility and implied variance. Out-of-the-money 25-delta puts and calls capture option-implied skew, while 10-delta puts and calls capture more extreme tail-risk pricing. The 25- and 10-delta put-call spreads measure asymmetry between downside and upside option prices, whereas smile-curvature variables compare the average implied volatility of out-of-the-money puts and calls with at-the-money implied volatility. Monthly predictors are constructed both from end-of-month option quotes and from averages of daily option-implied signals within each month.

Let $IV_{t,\Delta}$ denote the implied volatility for an option with delta Δ in month t. The at-the-money implied volatility is defined as

$$ATMIV_t = IV_{t,0.50},$$

and the corresponding implied variance is

$$IVAR_t = ATMIV_t^2.$$

Skew measures are constructed from out-of-the-money put and call implied volatilities:

$$Skew_{\Delta,t} = IV_{t,Put_\Delta} - IV_{t,Call_\Delta}.$$

Downside skew is defined as

$$DownsideSkew_{\Delta,t} = IV_{t,Put_\Delta} - ATMIV_t,$$

and curvature of the implied volatility smile is measured as

$$Curvature_{\Delta,t} = \frac{IV_{t,Put_\Delta} + IV_{t,Call_\Delta}}{2} - ATMIV_t.$$

Dispersion is constructed as the cross-sectional variation in implied volatility across contracts:

$$Disp_t = \frac{1}{N_t}\sum_{i=1}^{N_t} Dis\,p_{i,t}.$$

Variables are aggregated to the monthly frequency using either end-of-month values

$$X_t^{EOM} = X_{\tau(t)},$$

or monthly averages

$$X_t^{AVG} = \frac{1}{D_t}\sum_{d=1}^{D_t} X_{d,t}\,.$$

The variance risk premium is defined as

$$VRP_t^{expost} = IVAR_t - RV_{t+1},$$

which is not included due to look-ahead bias, however a real-time version is constructed as

$$VRP_t^{forecast} = IVAR_t - \widehat{RV}_{t+1|t}.$$

3.3. Macro-Financial Predictors

Macro-financial predictors are obtained from the Global Factor Data Library (Jensen et al., 2021). The dataset provides a large and systematically constructed set of variables capturing valuation, profitability, investment, risk, liquidity and funding conditions.

The predictors are observed at the monthly frequency and are aligned with the forecasting horizon so that only information available at the end of month t is used to forecast realized volatility in month $t+1$. This ensures consistency with the timing of realized volatility and option-implied variables.

Before estimation, the macro-financial variables are inspected for missing observations and standardized within each estimation window:

$$\tilde{x}_{j,t} = \frac{x_{j,t} - \mu_j}{\sigma_j}.$$

Given the high dimensionality and correlation structure of the dataset, dimension reduction and regularisation methods are employed in the empirical analysis to extract predictive information.

3.4. Final Forecasting Dataset

The final dataset consists of monthly observations containing the log realized variance target, HAR-style lagged realized-variance components, option-implied variables and macro-financial predictors. All predictors are aligned so that only information available at the end of month t is used to forecast realized variance in month $t+1$. This timing convention ensures a genuine out-of-sample forecasting setup and avoids look-ahead bias.

Since the models are estimated using log realized variance, forecasts are retransformed to variance units before evaluation. A direct exponential retransformation,

$$\exp\left(\hat{y}_{t+1|t}\right),$$

is generally downward biased for the conditional mean because of Jensen's inequality. The main results therefore use the lognormal correction,

$$\widehat{RV}_{t+1|t} = \exp\left(\hat{y}_{t+1|t} + \frac{1}{2}\hat{\sigma}^2_{\varepsilon,t}\right),$$

where $\hat{\sigma}^2_{\varepsilon,t}$ is estimated from residuals within the rolling estimation window. This places log-based forecasts and variance-based evaluation measures on a common scale and follows the retransformation-bias literature for exponentially transformed forecasts (Demetrescu et al., 2020).

The final common forecasting sample is determined by the intersection of the realized-variance target, HAR variables, option-implied predictors and macro-financial predictors. The main out-of-sample evaluation uses forecast origins from February 2011 to December 2023, corresponding to target months from March 2011 to January 2024 and $N_{OOS} = 155$ monthly forecasts. Within this common sample, all required HAR, option-implied and macro-financial predictors are available, and no months are skipped. The endpoint is determined by the macro-financial data, where the monthly market factor is available through December 2023. Although HAR-only and option-only models can be extended further, all main-table results are reported on the common sample to ensure comparability across information sets.

4. Methodology

4.1. Forecasting Framework

The empirical analysis is conducted in a pseudo out-of-sample rolling-window framework. At each forecast origin t, the objective is to forecast next-month log realized variance,

$$y_{t+1} = \log(RV_{t+1}),$$

using only information available at the end of month t. This timing convention ensures that historical realized-variance components, option-implied variables and macro-financial predictors enter the forecasting exercise without look-ahead bias.

The main results use a rolling estimation window of $W = 120$ monthly observations, corresponding to a ten-year training sample. At each forecast origin, all model parameters, predictor transformations, principal components, tuning choices and forecast combinations are estimated using only the most recent 120 months and then rolled forward by one month. This design follows Audrino and Chassot (2024), who emphasize that training-window length and re-estimation choices matter when comparing HAR benchmarks with machine learning volatility forecasts.

4.2. HAR

The benchmark model is the heterogeneous autoregressive model of realized volatility, HAR-RV (Corsi, 2009). Since the forecasting target is log realized variance, the monthly HAR specification is

$$y_{t+1} = \alpha + \beta_1 y_t + \beta_3 y_t^{(3)} + \beta_{12} y_t^{(12)} + \varepsilon_{t+1},$$

where

$$y_t = \log(RV_t),$$

and RV_t denotes monthly realized variance. The three HAR components capture information from the previous month, quarter and year, respectively. The model serves as the main historical-volatility benchmark and as the autoregressive component included in the richer option-implied and macro-financial specifications.

4.3. Machine Learning Forecasting Methods

Machine learning methods are included to handle high-dimensional, correlated and potentially nonlinear predictor sets. The purpose is to test whether regularisation, dimension reduction and nonlinear modelling can extract predictive information beyond the HAR benchmark, while recognising that flexible models do not necessarily dominate strong linear volatility models (Christensen et al., 2022; Branco et al., 2024; Gunnarsson et al., 2024; Audrino and Chassot, 2024).

Let X_t denote the predictor vector observed at the end of month t. All models are estimated as one-step-ahead forecasts,

$$y_{t+1} = f(X_t) + \varepsilon_{t+1},$$

where $y_{t+1} = \log(RV_{t+1})$.

4.3.1. Principal Components Regression

Principal Components Regression is used as a linear dimension-reduction method. In each estimation window, predictors are standardized and transformed into principal components, which summarize common variation in the predictor set. This is useful for correlated option-implied and macro-financial variables.

Let PC_t denote the retained components extracted from X_t. The forecasting equation is

$$y_{t+1} = \alpha + \beta' PC_t + \varepsilon_{t+1}.$$

The number of components is selected to explain at least 70% of total predictor variation, and both the PCA transformation and regression coefficients are re-estimated at each forecast origin. This follows the factor-forecasting literature for large macro-financial datasets (Stock and Watson, 2002; Ludvigson and Ng, 2007; McCracken and Ng, 2016).

4.3.2. Elastic Net

The Elastic Net provides a regularized linear forecasting model that combines Ridge and Lasso penalties. This makes it well suited to settings with many correlated predictors, since it both shrinks coefficients and allows for variable selection. The model is estimated as

$$y_{t+1} = \alpha + X_t' \beta + \varepsilon_{t+1}.$$

The coefficient vector is obtained by minimizing

$$\sum_t (y_{t+1} - \alpha - X_t'\beta)^2 + \lambda \left[\frac{1-\eta}{2} \sum_j \beta_j^2 + \eta \sum_j |\beta_j| \right],$$

where λ controls the overall degree of regularization and η determines the balance between Ridge and Lasso shrinkage. In the empirical implementation, $\eta = 0.5$, giving equal weight to the two penalty components, while λ is selected by cross-validation within each estimation window. The model is re-estimated at each monthly forecast origin. Elastic Net is included as a regularized linear model for high-dimensional and collinear predictor sets (Zou and Hastie, 2005; Gu et al., 2020; Christensen et al., 2022).

4.3.3. Random Forest

Random Forest is included to capture nonlinear relationships and interaction effects between predictors. The method constructs an ensemble of regression trees, where each tree is estimated on a bootstrap sample of the training data and each split considers only a random subset of predictors. Forecasts are obtained by averaging across the individual trees,

$$\hat{y}_{t+1} = \frac{1}{B}\sum_{b=1}^{B} T_b\,(X_t),$$

where $T_b(\cdot)$ denotes the prediction from tree b, and B is the number of trees.

In the empirical implementation, each forest consists of 100 trees. At each split, the number of candidate predictors is set equal to the square root of the total number of available predictors, and tuning is based on out-of-bag prediction errors. Model complexity is controlled through the tree-growing procedure and tuning parameters selected using out-of-bag prediction errors. The model is re-estimated at each monthly forecast origin. Its inclusion allows the analysis to test whether nonlinearities and interactions between historical, option-implied and macro-financial predictors improve forecasts beyond linear methods (Breiman, 2001; Gu et al., 2020; Christensen et al., 2022).

4.3.4. Neural Network

A feedforward neural network is used as a flexible nonlinear forecasting model. Neural networks can approximate complex nonlinear relationships between predictors and future volatility, but they are also prone to overfitting in relatively small samples. To reduce dimensionality, the neural network is estimated using a limited number of principal components extracted from the predictor set.

Let PC_t denote the retained principal components. The forecasting model is

$$y_{t+1} = f(PC_t) + \varepsilon_{t+1},$$

where $f(\cdot)$ is approximated by a single-hidden-layer neural network,

$$f(PC_t) = \alpha_0 + \sum_{h=1}^{H} \alpha_h\, g(\delta_{h0} + \theta_h' PC_t).$$

Here, H denotes the number of hidden units and $g(\cdot)$ is the activation function. In the empirical implementation, the number of principal components is capped at six, the hidden layer contains three units, the weight decay parameter is set to 0.1, and estimation is run for a maximum of 300 iterations. To reduce sensitivity to random initialization, five networks are estimated with different starting values and the final forecast is obtained by averaging their predictions. The PCA transformation and neural network are re-estimated at each monthly forecast origin.

Neural networks are included to examine whether more flexible nonlinear approximations can extract predictive information from the combined information set beyond linear regularization and tree-based methods (Hornik et al., 1989; Christensen et al., 2022; Ge et al., 2022; Gunnarsson et al., 2024).

4.3.5. Information Sets and Comparability

For each machine learning method, forecasts are generated using separate predictor blocks: option-implied variables, macro-financial predictors and the combined information set. Historical HAR components are included as the baseline volatility information set where relevant, so that the empirical analysis evaluates whether option-implied and macro-financial variables provide incremental predictive value beyond past realized volatility. This design keeps the comparison consistent across methods and allows forecast performance to be attributed both to the forecasting algorithm and to the information set used.

4.4. Forecast Combinations

Forecast combinations are included to reduce model-selection risk and exploit complementary information across models. Equal-weight combinations are formed across information sets within each machine learning method, across machine learning methods within each information set, and across all individual forecasts. This follows the forecast-combination literature, where pooling forecasts can improve out-of-sample performance relative to selecting a single model ex ante (Granger and Ramanathan, 1984; Hansen, 2008; Wang et al., 2022).

Stacked combinations use the individual forecasts as inputs to a second-stage model,

$$\hat{y}_{t+1|t}^{stack} = g(Z_t),$$

where $Z_t$denotes the vector of forecasts available at forecast origin t. This approach is related to stacked regression and stacked generalization, where a meta-learner combines base-model predictions to improve accuracy (Breiman, 1996; Naimi and

Balzer, 2018). In this application, stacking is relevant because HAR, option-based, macro-based and machine-learning forecasts may capture different components of future realized volatility.

Elastic Net and Random Forest are used as stacking models. Elastic Net can shrink weak forecasts toward zero, while Random Forest allows nonlinear forecast interactions and provides variable-importance diagnostics. To avoid look-ahead bias, the second-stage model is re-estimated recursively using only past forecast-outcome pairs, $\{Z_s, y_{s+1}\}_{s<t}$, before being applied to the forecasts for $t+1$. Thus, stacked forecasts are genuine out-of-sample combinations rather than ex-post fitted averages.

4.5. Model Comparison and Forecast Evaluation

Forecast performance is evaluated out of sample by comparing predicted and realized next-month variance. Since the models are estimated in log realized variance, forecasts are retransformed to variance units before evaluation. The first evaluation measure is the out-of-sample R^2, defined relative to the HAR benchmark:

$$R^2_{OOS} = 1 - \frac{\sum_t (RV_{t+1} - \widehat{RV}^{i}_{t+1|t})^2}{\sum_t (RV_{t+1} - \widehat{RV}^{HAR}_{t+1|t})^2}.$$

A positive value indicates that model i reduces squared prediction errors relative to the HAR benchmark. This measure follows the predictive-regression literature, where forecast performance is assessed relative to a benchmark model (Campbell and Thompson, 2008).

The second evaluation measure is QLIKE,

$$QLIKE_{i,t+1} = \frac{RV_{t+1}}{\widehat{RV}^{i}_{t+1|t}} - \log\left(\frac{RV_{t+1}}{\widehat{RV}^{i}_{t+1|t}}\right) - 1.$$

A lower QLIKE indicates a more accurate variance forecast. QLIKE is widely used in volatility forecast evaluation because it is robust to noise in volatility proxies under suitable conditions (Patton, 2011). Reporting both R^2_{OOS} and QLIKE reduces dependence on a single loss function.

Statistical significance is assessed using one-sided Diebold-Mariano tests against the OLS HAR benchmark (Diebold and Mariano, 1995). For each competing model i, the loss differential is defined as

$$d_{i,t+1} = L_{HAR,t+1} - L_{i,t+1},$$

so that a positive mean loss differential indicates lower average loss than HAR. Squared-error loss is used for the RMSE and R^2_{OOS} columns, while QLIKE loss is used for the QLIKE gain column. The test therefore evaluates whether the observed forecast gains are statistically larger than zero, rather than only numerically positive.

In addition to statistical forecast evaluation, economic significance is assessed using a stylised variance-risk-premium timing exercise. The signal is defined as implied variance minus forecasted realized variance, and is used to scale short-volatility exposure. Because this exercise requires additional assumptions about scaling, transaction costs and portfolio constraints, the full economic-value results are reported in the appendix.

5. Empirical Results

This section presents the out-of-sample forecasting results. The models are evaluated relative to the HAR benchmark using R^2_{OOS}, QLIKE gains and cumulative forecast-error gains. The main paper focuses on the central patterns, while full model rankings, robustness checks, variable-importance diagnostics and economic-value results are reported in the appendix.

5.1. Main Forecast Comparison

Table 1 reports the main out-of-sample forecast comparison. The strongest improvements relative to the HAR benchmark are obtained by models that include option-implied information, especially Elastic Net HAR + Option, Random Forest HAR + Option and the HAR + Option equal-weighted combination. Forecast combinations are also competitive, suggesting that pooling across models and information sets reduces model-specific noise. Macro-financial predictors deliver more mixed results: some HAR + Macro specifications improve under R^2_{OOS}, but their QLIKE performance is less stable, and HAR + Macro + Option models do not clearly dominate the simpler HAR + Option specifications.

The strong R^2_{OOS} performance of the HAR-only Random Forest indicates that nonlinear transformations of historical volatility also improve squared-error performance. However, its near-zero QLIKE gain suggests that this improvement is less robust across loss functions than the gains from option-implied specifications.

Table 1. Out-of-Sample Forecast Performance Relative to the HAR Benchmark

Model	Info set	R^2_{OOS} rank	QLIKE rank	M	N_{OOS}	RMSE gain	R^2_{OOS}	QLIKE gain	R^2_{OOS} 60m	R^2_{OOS} 180m
Elastic Net	HAR+Option	1	1	1	155	0.076	0.878	0.128*	0.698	0.558*
Equal Weight	HAR+Option	2	5	4	155	0.068	0.827	0.087	0.852*	0.418
Random Forest	HAR+Option	3	7	1	155	0.067	0.823	0.066*	0.949	0.487*
Stacked Random Forest	Multiple	4	15	16	155	0.066	0.815	0.016	0.943	0.411
Random Forest EW	Multiple	5	14	4	155	0.065	0.806	0.031	0.951	0.445
Random Forest	HAR+Macro+Option	6	12	1	155	0.065	0.804	0.037	0.946	0.452
Random Forest	HAR	7	24	1	155	0.062	0.786	-0.033	0.883	0.325
Random Forest	HAR+Macro	8	26	1	155	0.059	0.753	-0.129	0.880	0.260
Stacked Elastic Net	Multiple	9	9	16	155	0.057	0.745	0.063*	0.830	0.494*
Neural Network	HAR+Macro+Option	10	21	1	155	0.057	0.741	-0.019	0.941	0.203
Equal Weight	HAR+Macro+Option	11	2	4	155	0.054	0.716	0.098*	0.871*	0.420*
Neural Network EW	Multiple	12	11	4	155	0.051	0.689	0.045	0.942	0.286
Neural Network	HAR+Macro	13	27	1	155	0.049	0.665	-0.267	0.888	-0.155
Equal Weight	Multiple	14	4	16	155	0.047	0.648	0.088**	0.814*	0.424*
Equal Weight	HAR+Macro	15	23	4	155	0.038	0.548	-0.027	0.606*	0.199
Elastic Net EW	Multiple	16	3	4	155	0.038	0.548	0.089*	0.632	0.436*
PCA	HAR+Option	17	6	1	155	0.026	0.400	0.075	0.266*	0.036
Elastic Net	HAR	18	20	1	155	0.024	0.377	-0.014	0.657	0.071
Neural Network	HAR+Option	19	28	1	155	0.024	0.366	-1.967	0.851	-1.594
Elastic Net	HAR+Macro+Option	20	13	1	155	0.023	0.353	0.037	0.714	0.363*
Equal Weight	HAR	21	16	4	155	0.012	0.191	0.003	0.740	0.282
PCA	HAR+Macro+Option	22	8	1	155	0.011	0.183	0.064	0.208	0.014
PCA EW	Multiple	23	10	4	155	0.004*	0.064*	0.062*	-0.017	0.006
Elastic Net	HAR+Macro	24	19	1	155	0.000	0.001	-0.003	-0.081	0.012*
OLS HAR	HAR	25	17	1	155	0.000	0.000	0.000	0.000	0.000
PCA	HAR	25	17	1	155	0.000	0.000	0.000	0.000	0.000
PCA	HAR+Macro	27	22	1	155	-0.026	-0.494	-0.022	-0.786	-0.076
Neural Network	HAR	28	25	1	155	-0.080	-1.862	-0.044	0.739	0.340

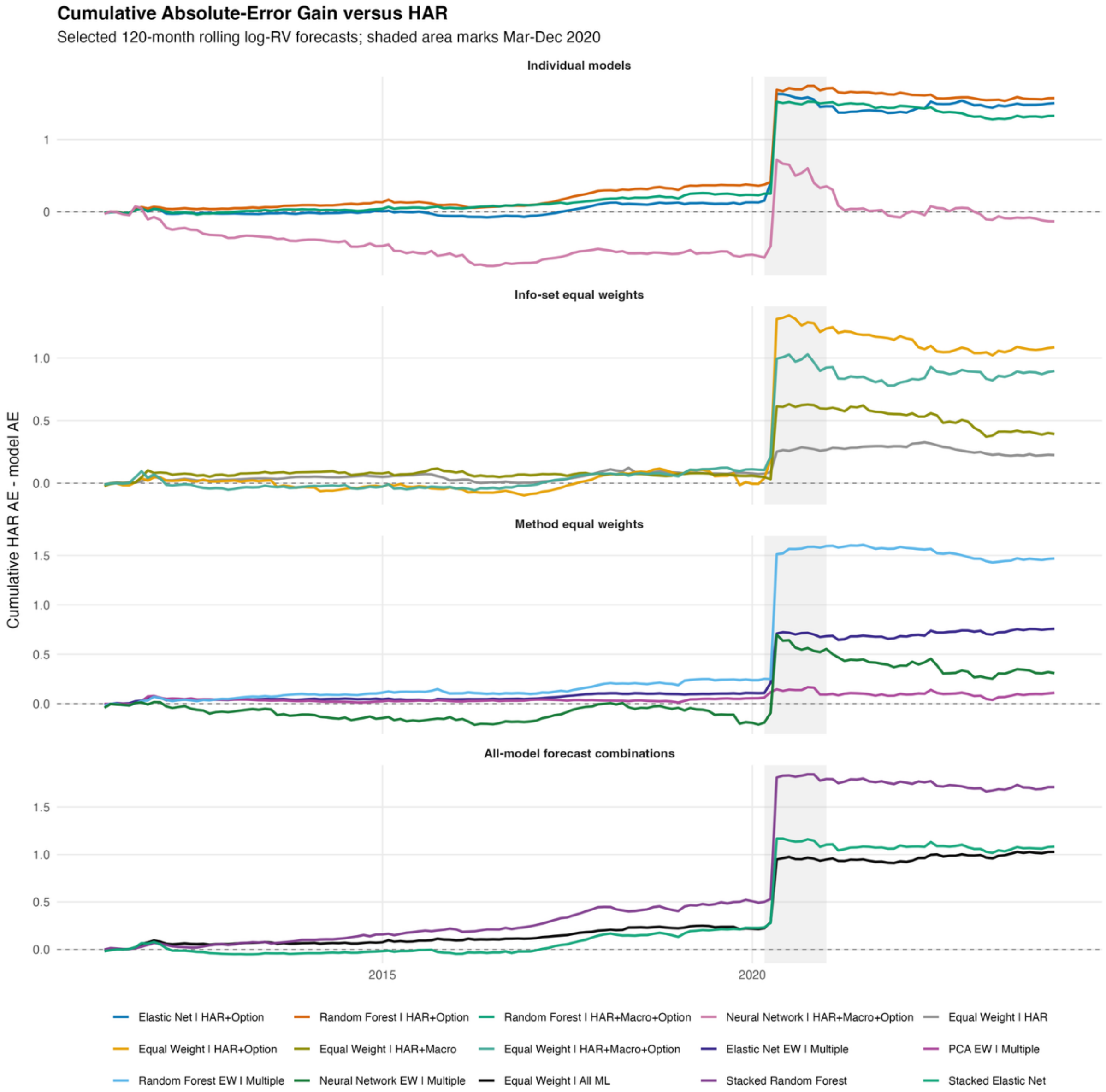
Notes: Rows are sorted by the 120-month out-of-sample R^2_{OOS} rank. The common evaluation sample consists of target months March 2011 to January 2024, using forecast origins February 2011 to December 2023. The benchmark is the OLS HAR model. N_{OOS} is the number of out-of-sample forecasts and $M$is the number of forecasts in a combination. RMSE gain, QLIKE gain and R^2_{OOS} are measured relative to HAR, so positive values indicate better performance; positive RMSE gain means lower RMSE than HAR. The 60-month and 180-month columns report robustness to alternative rolling-window lengths. Stars denote one-sided Diebold-Mariano tests against OLS HAR: RMSE/R^2 columns use squared-error loss for the relevant window, while QLIKE gain uses corrected rolling-120 QLIKE loss. $^{**}p < 0.05$, $^{*}p < 0.10$.

5.2. Forecast Gains over Time

Figure 1 shows that option-based models generate modest gains before 2020 and much larger gains during the COVID volatility shock. The strongest individual models are Elastic Net HAR + Option and Random Forest HAR + Option, while forecast combinations also accumulate positive gains after 2020. This confirms the main table result: the most stable forecast improvements come from option-implied information, especially when combined with regularised, tree-based or combination methods.

This suggests that option-implied information is particularly valuable in stress periods, while pre-2020 gains are more modest.

Figure 1. Cumulative Forecast-Error Gains Relative to the HAR Benchmark



Notes: The figure shows cumulative absolute-error gains relative to the HAR benchmark. Positive values indicate that a model has accumulated lower forecast errors than HAR. Gains are computed at each forecast origin as the HAR absolute error minus the competing model's absolute error.

5.3. Forecast Diagnostics, Economic Value and Robustness

Forecast diagnostics support the interpretation that the gains are mainly driven by option-implied information. In Elastic Net and Random Forest models, the most important additional predictors are concentrated in the option block, including at-the-money implied volatility, 25-delta implied volatilities, implied variance and dispersion measures. Stacked forecast diagnostics point in the same direction, with the largest contributions typically coming from HAR + Option or HAR + Macro + Option forecasts. This suggests that the improvements are driven by information in the implied volatility surface rather than by model complexity alone.

Economic significance is evaluated through a stylised variance-risk-premium timing exercise, where implied variance minus forecasted realized variance determines short-volatility exposure. The results are broadly consistent with the statistical findings for moderate risk aversion, but they depend on scaling, transaction costs, risk aversion and the short-volatility constraint. The full Sharpe ratios, certainty-equivalent gains, turnover measures and expanding CEQ results are therefore reported in the appendix.

The main conclusions are also robust to alternative rolling-window lengths. Option-based models and forecast combinations remain among the strongest specifications, especially Elastic Net HAR + Option, Random Forest HAR + Option and HAR + Option equal-weighted combinations. Full diagnostic, economic-value and robustness tables are reported in the appendix.

6. Discussion

6.1. Option-Implied Information and Forecast Gains

The central finding is that option-implied variables provide the clearest incremental forecasting value. This is consistent with Christensen and Prabhala (1998), who show that implied volatility contains information about future realized volatility beyond historical measures. Busch et al. (2011) provide similar evidence across several asset classes, suggesting that option markets aggregate forward-looking information relevant for future volatility.

The results also align with more recent work on HAR-type models and option-derived predictors. Kambouroudis et al. (2021) show that augmenting HAR models with implied volatility improves realized volatility forecasts across international stock indices. Michael et al. (2025) further emphasize that richer measures extracted from the implied volatility surface can improve forecast accuracy. The present results support this broader view, since the strongest models are not based on a single implied volatility measure, but on a wider option-implied information set including volatility levels, skew, curvature, dispersion and variance risk premia.

Economically, this finding is intuitive. Option prices reflect market expectations under the risk-neutral measure and incorporate compensation for volatility and tail risk. During periods of market stress, such as the 2020 volatility shock, this forward-looking information appears especially valuable because option prices adjust quickly to changing uncertainty. This explains why the cumulative forecast gains increase sharply around crisis periods.

6.2. Machine Learning and Model Specification

The results suggest that machine learning methods are useful, but not because flexible models automatically dominate HAR benchmarks. The strongest performance comes from Elastic Net and Random Forest models applied to option-implied predictors. This indicates that the gains arise from combining economically meaningful information with methods that can handle high-dimensionality, multicollinearity and nonlinearities.

This interpretation is consistent with Christensen et al. (2022), who show that machine learning methods can be competitive with HAR-type models when additional predictors are available. Gunnarsson et al. (2024) similarly emphasize that machine learning can perform well in volatility forecasting, while traditional econometric models remain important benchmarks.

At the same time, the results support the more cautious conclusions of Branco et al. (2024) and Audrino and Chassot (2024). Flexible methods do not automatically outperform simpler models, and well-specified linear benchmarks remain difficult to beat. In the present analysis, Neural Network specifications are less stable than Elastic Net and Random Forest, suggesting that model complexity can increase overfitting risk in a monthly sample. The main implication is therefore that machine learning adds value when it is regularised, robust and paired with informative predictors.

6.3. Macro-Financial Predictors and Forecast Combinations

Macro-financial predictors are theoretically relevant but empirically less robust than option-implied variables. This contrasts with evidence from Ludvigson and Ng (2007), Engle et al. (2013) and Nonejad (2017), who show that macro-financial variables can help explain risk, returns and volatility. One explanation is that macro-financial variables evolve slowly and may be more informative at longer horizons, while option prices already incorporate expectations about macroeconomic risk, uncertainty and downside protection.

Forecast combinations are a more robust finding. Both equal-weighted and stacked combinations perform competitively, suggesting that pooling forecasts helps reduce model-specific noise and avoids relying on one model selected ex ante. This is consistent with Timmermann (2006) and Rapach et al. (2009), who show that combinations can improve out-of-sample performance when predictive relationships are unstable. In this setting, the result is economically intuitive: HAR captures persistence, option-implied variables capture forward-looking market expectations, macro-financial variables capture broader conditions, and machine learning methods capture regularised or nonlinear relationships. The strong performance of combinations therefore supports the view that volatility forecasting benefits from pooling complementary signals, even when option-implied information remains the dominant source of incremental predictive power.

6.4. Economic Value and Limitations

The economic-value results suggest that the forecast improvements are not only statistical, since option-based and combination models generate positive certainty-equivalent gains relative to HAR in the variance-risk-premium timing strategy. However, these results should be interpreted cautiously because they depend on assumptions about scaling, transaction costs and the short-volatility constraint. Moreover, lower average losses do not always imply statistically significant dominance, as emphasized by the Diebold-Mariano framework for equal predictive accuracy (Diebold and Mariano, 1995). Using both R^2_{OOS} and QLIKE helps address this issue: Campbell and Thompson (2008) motivate benchmark-relative out-of-sample evaluation, while Patton (2011) shows that QLIKE is appropriate for volatility forecast evaluation under noisy volatility proxies.

7. Conclusion

This paper examines whether option-implied and macro-financial predictors improve monthly forecasts of S&P 500 realized variance beyond historical volatility dynamics. The empirical framework compares a HAR benchmark with machine learning models estimated on option-implied variables, macro-financial predictors and combined information sets, evaluated out of sample using R^2_{OOS} relative to HAR and QLIKE loss.

The main finding is that option-implied variables provide the clearest incremental forecasting value. HAR + Option specifications perform strongly across the main evaluation criteria, suggesting that option markets contain forward-looking information not fully captured by past realized volatility. Machine learning methods add value mainly when applied to informative predictor sets, especially through regularisation and tree-based methods rather than complexity alone. Macro-financial predictors are theoretically relevant but less robust once option-implied information is included, while forecast combinations improve robustness by reducing model-specific noise.

Overall, the evidence suggests that monthly volatility forecasts benefit most from combining historical volatility dynamics with option-implied information. Future work could examine alternative horizons, indices, realized-variance measures and economic-value specifications.

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9. Appendix

This appendix reports the full economic-value exercise and the forecast-diagnostic tables that support the main results. The material is organised to match the structure of the paper. Appendix A documents the variance-risk-premium timing strategy, Appendix B reports model-level diagnostics, and Appendix C reports stacked forecast diagnostics. All results are computed using the main forecast panel: log realized volatility, a rolling 120-month estimation window, end-of-month implied variance, normalized variance-payoff units and a transaction cost of 0.001 per unit of monthly turnover.

9.1. Appendix A. Economic-Value Timing Strategy

The economic-value exercise evaluates whether differences in realized-variance forecasts translate into a stylised variance-risk-premium timing signal. At the end of month t, implied variance is observable from option prices, while realized variance for month $t+1$is still unknown. Each model forecast is therefore converted to variance units and compared with the option-implied variance observed at the same forecast origin.

The ex-ante variance-risk-premium signal is defined as

$$VRP_t^{signal} = IVAR_t - \widehat{RV}_{t+1|t},$$

where $IVAR_t$is the implied variance observed at the end of month t, and $\widehat{RV}_{t+1|t}$is the model forecast of realized variance for month $t+1$.

Since the forecasting models are estimated in log realized variance, the forecast is first transformed back to variance units before entering the trading rule:

$$\widehat{RV}_{t+1|t} = \exp\left(\hat{y}_{t+1|t} + \frac{1}{2}\hat{\sigma}_{\varepsilon,t}^2\right).$$

The signal is scaled by its own rolling volatility, computed using only past signal observations, and then constrained to be short-only:

$$w_t = \max\left(0, \frac{VRP_t^{signal}}{\hat{\sigma}_{VRP,t}}\right).$$

The realized normalized payoff is observed only after month $t+1$:

$$R_{t+1}^{gross} = w_t \frac{IVAR_t - RV_{t+1}}{IVAR_t}.$$

Transaction costs are deducted according to monthly turnover:

$$R_{t+1}^{net} = R_{t+1}^{gross} - c \mid w_t - w_{t-1} \mid .$$

Certainty-equivalent performance is calculated from monthly net returns and reported as gains relative to the HAR scaled short-only benchmark:

$$CEQ_i = 12\left(\bar{R}_i^{net} - \frac{\gamma}{2} Var(R_i^{net})\right).$$

This timing convention is important because the signal is entirely ex ante: $IVAR_t$and the model forecast are known at the end of month t, while RV_{t+1}enters only when the payoff is realized. Hence, the strategy uses the forecast to decide whether the expected volatility premium is attractive enough to justify short-volatility exposure.

Table A1. Timing and Payoff Assumptions

Object	Timing / calculation
Implied variance	Observed at the end of month t using implied_var_eom.
Forecast variance	One-month-ahead realized-variance forecast for month t+1, converted to variance units before comparison.
Rolling signal volatility	Computed only from past VRP signals dated strictly before month t.
Short-volatility payoff	Observed after month t+1 as implied variance at t minus realized variance in t+1.
Normalized gross return	Position times the payoff, divided by implied variance at t.
Net return	Gross return minus 0.001 times monthly absolute position change.

Notes: The strategy uses only information available at the end of month t when forming the position. Realized variance for month $t+1$ is used only to calculate the ex-post payoff.

Table A2. Specification Types

Specification type	Meaning
ind.	One individual machine-learning forecast specification.
info EW	Equal-weight average across machine-learning methods within one information set.
method EW	Equal-weight average across information sets within one model family.
all EW	Equal-weight average across the broader machine-learning forecast panel.
stacked	Meta-model forecast combination estimated from the individual forecast panel.
naive	Simple non-machine-learning benchmark forecast.

Notes: Equal-weight and stacked combinations use the individual forecasts from the main forecast panel.

Table A3. Column Definitions for the Economic-Value Tables

Column	Calculation / meaning
Rank	Overall rank by annualized net Sharpe ratio.
Model	Forecast model or combination method.
Info set	Predictor information set used by the forecast.
Type	Forecast or combination design.
Sharpe	sqrt(12) times mean net return divided by standard deviation of net return.
Ann. mean	12 times average monthly net return.
Ann. Vol	sqrt(12) times monthly net-return standard deviation.
CEQ gamma	Annualized certainty-equivalent gain relative to HAR for risk aversion gamma.
Traded	Fraction of months with positive short-volatility exposure.
Turnover	Average monthly absolute position change.

Notes: Return moments are annualized from monthly net returns. CEQ gains are measured relative to the HAR scaled short-only benchmark under the same trading rule.

Table A4. Scaled Short-Only VRP Strategy: Core Performance Ranking

Rank	Model	Info set	Type	Sharpe	Ann. mean	Ann. vol	Traded	Turnover
1	Elastic Net	HAR+Option	ind.	2.659	3.038	1.143	0.845	0.187
2	Random Forest	HAR+Macro+Option	ind.	2.368	2.716	1.147	0.826	0.218
3	Random Forest	HAR+Option	ind.	2.367	2.831	1.196	0.845	0.188
4	Stacked Random Forest	Multiple	stacked	2.326	2.674	1.150	0.839	0.201
5	Stacked Elastic Net	Multiple	stacked	2.307	2.521	1.093	0.839	0.226
6	Equal Weight	HAR+Macro+Option	info EW	2.275	2.450	1.077	0.806	0.253
7	Neural Network	HAR+Macro+Option	ind.	2.254	2.341	1.038	0.723	0.280
8	Random Forest EW	Multiple	method EW	2.247	2.449	1.090	0.832	0.220
9	Equal Weight	HAR+Option	info EW	2.116	2.501	1.182	0.806	0.246
10	Random Forest	HAR	ind.	2.084	2.191	1.051	0.826	0.232
11	Neural Network	HAR+Macro	ind.	2.059	1.875	0.910	0.697	0.267
12	Random Forest	HAR+Macro	ind.	2.026	2.109	1.041	0.735	0.229
13	Neural Network EW	Multiple	method EW	1.957	2.018	1.031	0.761	0.284
14	All ML forecasts EW	Multiple	all EW	1.920	1.844	0.960	0.819	0.233
15	Equal Weight	HAR+Macro	info EW	1.821	1.590	0.873	0.781	0.230
16	Elastic Net EW	Multiple	method EW	1.725	1.579	0.915	0.826	0.206
17	PCA	HAR+Option	ind.	1.621	1.439	0.887	0.800	0.185
18	Neural Network	HAR+Option	ind.	1.617	1.523	0.942	0.697	0.269
19	PCA	HAR+Macro+Option	ind.	1.585	1.475	0.930	0.794	0.184
20	Elastic Net	HAR+Macro+Option	ind.	1.552	1.525	0.983	0.813	0.210
21	Elastic Net	HAR	ind.	1.546	1.264	0.818	0.781	0.200
22	PCA EW	Multiple	method EW	1.459	1.244	0.853	0.794	0.175
23	Equal Weight	HAR	info EW	1.451	1.212	0.835	0.819	0.195
24	Elastic Net	HAR+Macro	ind.	1.405	1.139	0.810	0.813	0.179
25	HAR	Short-only	HAR	1.391	1.106	0.795	0.800	0.177
26	PCA	HAR	ind.	1.391	1.106	0.795	0.800	0.177
27	PCA	HAR+Macro	ind.	1.331	1.091	0.820	0.794	0.172
28	Neural Network	HAR	ind.	1.237	1.050	0.849	0.787	0.194
29	Historical Mean RV	Historical RV	naive	1.005	0.715	0.712	0.213	0.117
30	IV as RV Forecast	Option	naive	NA	0.000	0.000	0.000	0.000

Notes: Rows are sorted by annualized net Sharpe ratio. The strategy uses implied_var_eom, normalized variance-payoff units and a cost of 0.001 per unit of monthly turnover. The values are not literal fully implemented option-portfolio returns.

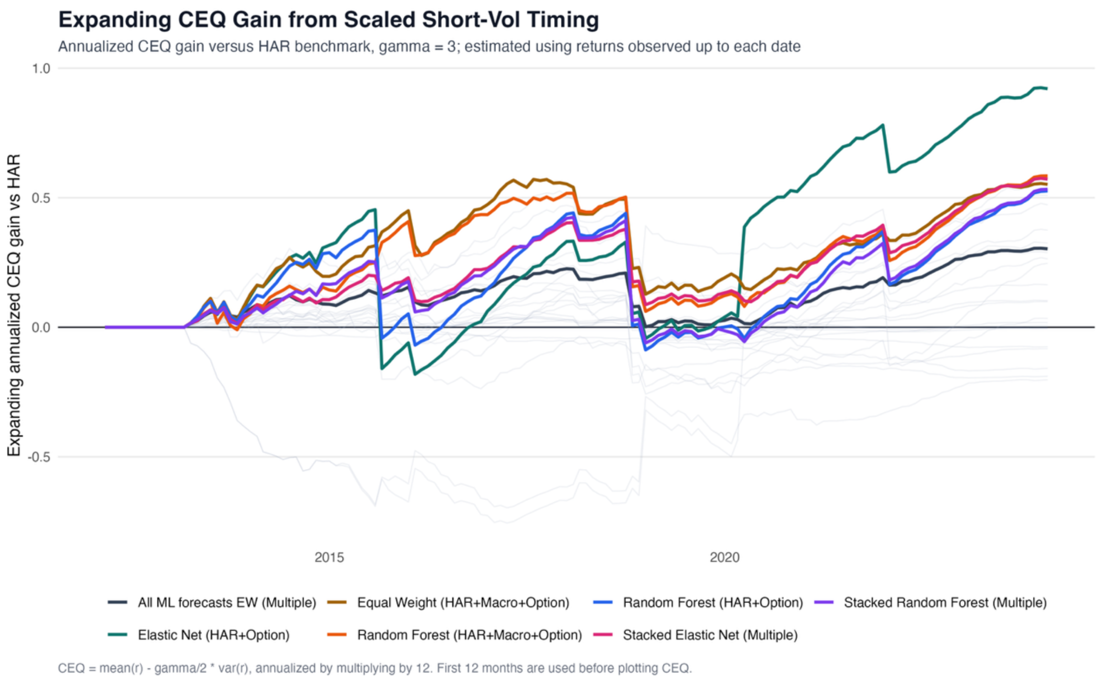
Table A5. Scaled Short-Only VRP Strategy: CEQ Gains Relative to HAR

Rank	Model	Info set	CEQ gamma=3	CEQ gamma=5
1	Elastic Net	HAR+Option	0.921	0.247
2	Random Forest	HAR+Macro+Option	0.584	-0.100
3	Random Forest	HAR+Option	0.526	-0.273
4	Stacked Random Forest	Multiple	0.533	-0.157
5	Stacked Elastic Net	Multiple	0.572	0.009
6	Equal Weight	HAR+Macro+Option	0.552	0.023
7	Neural Network	HAR+Macro+Option	0.565	0.118
8	Random Forest EW	Multiple	0.509	-0.047
9	Equal Weight	HAR+Option	0.247	-0.518
10	Random Forest	HAR	0.375	-0.099
11	Neural Network	HAR+Macro	0.473	0.275
12	Random Forest	HAR+Macro	0.325	-0.127
13	Neural Network EW	Multiple	0.264	-0.168

Rank	Model	Info set	CEQ gamma=3	CEQ gamma=5
14	All ML forecasts EW	Multiple	0.303	0.013
15	Equal Weight	HAR+Macro	0.288	0.158
16	Elastic Net EW	Multiple	0.164	-0.042
17	PCA	HAR+Option	0.099	-0.057
18	Neural Network	HAR+Option	0.034	-0.222
19	PCA	HAR+Macro+Option	0.018	-0.216
20	Elastic Net	HAR+Macro+Option	-0.082	-0.417
21	Elastic Net	HAR	0.103	0.066
22	PCA EW	Multiple	-0.005	-0.100
23	Equal Weight	HAR	0.006	-0.060
24	Elastic Net	HAR+Macro	-0.005	-0.030
25	HAR	Short-only	0.000	0.000
26	PCA	HAR	0.000	0.000
27	PCA	HAR+Macro	-0.075	-0.115
28	Neural Network	HAR	-0.189	-0.278
29	Historical Mean RV	Historical RV	-0.203	-0.078
30	IV as RV Forecast	Option	-0.158	0.473

Notes: CEQ gains are annualized and measured relative to the HAR scaled short-only benchmark. Positive values indicate higher certainty-equivalent performance than HAR for the same trading rule and risk-aversion parameter.

Figure A.1. Expanding Certainty-Equivalent Gains from Scaled Short-Volatility Timing



Notes: The figure reports expanding annualized certainty-equivalent (CEQ) gains relative to the HAR benchmark for the variance-risk-premium timing strategy. Gains are computed using risk aversion $\gamma = 3$ and only returns observed up to each date. Positive values indicate higher economic value than the HAR benchmark. Coloured lines show selected models and forecast combinations, while thin grey lines show the remaining specifications. The horizontal zero line corresponds to the HAR benchmark.

9.2. Appendix B. Individual Model Diagnostics

This appendix reports model-level diagnostics for the main 120-month rolling log-RV specification. Elastic Net diagnostics are reported as non-zero coefficient frequency across rolling refits. Random Forest diagnostics are reported using impurity importance and top-five frequency across refits. The purpose is to examine whether the strongest forecasting results are linked to economically interpretable predictors rather than only to model complexity. The main forecasting exercise uses 155 out-of-sample observations because stacked forecasts require an additional second-stage estimation history. The diagnostic tables use all available rolling refits and are therefore intended as interpretive diagnostics rather than direct performance comparisons.

Table B1. Elastic Net HAR+Option Variable Selection

Variable	Group	Selected refits	Selection rate	Mean |coef.|	Median |coef.|	Latest |coef.|
realized vol	HAR	237	1.00000	0.19059	0.18944	0.25309
3-month mean of realized vol	HAR	237	1.00000	0.11723	0.09296	0.01979
12-month mean of realized vol	HAR	237	1.00000	0.05693	0.02913	0.00817
call 25 iv eom	Option	205	0.86498	0.12917	0.12546	0.16520
atm iv eom	Option	203	0.85654	0.10827	0.09894	0.04146
put 25 dispersion eom	Option	187	0.78903	0.10118	0.08954	0.00912
implied var month avg	Option	185	0.78059	0.18085	0.19254	0.15848
mean dispersion eom	Option	177	0.74684	0.03250	0.02746	0.03458
smile curvature 25 month avg	Option	174	0.73418	0.05113	0.05011	0.00252
put 25 iv eom	Option	169	0.71308	0.05542	0.04513	0.03351
atm dispersion eom	Option	161	0.67932	0.10111	0.09047	0.01958
smile curvature 25 eom	Option	161	0.67932	0.06587	0.06243	0.00363

Notes: Variables are sorted by selection rate and then by mean absolute coefficient. The three HAR variables are included by construction, while the option variables are selected by the penalty. The table reports diagnostics across rolling refits.

Table B2. Random Forest HAR+Option Variable Importance

Variable	Group	Refits	Top-5 refits	Top-5 rate	Mean imp.	Median imp.	Latest imp.	Mean rank
put 25 iv eom	Option	217	216	0.99539	1.91174	1.99253	1.90260	2.02765
atm iv eom	Option	217	216	0.99539	1.80977	1.75651	2.73043	2.73733
implied var eom	Option	217	201	0.92627	1.84028	1.70247	2.65082	2.62673
call 25 iv eom	Option	217	198	0.91244	1.72806	1.53216	2.36912	3.51613
realized vol	HAR	217	77	0.35484	1.25963	1.28504	1.65487	6.74194
atm dispersion eom	Option	217	56	0.25806	1.04147	0.94474	0.54288	8.38249
implied var month avg	Option	217	35	0.16129	1.00304	0.91084	1.47811	9.35945
skew 10 eom	Option	217	27	0.12442	0.89870	0.96093	0.51149	10.43318
downside skew 10 eom	Option	217	14	0.06452	0.60744	0.55739	0.30518	15.57143
vrp forward eom	Option	217	10	0.04608	0.77477	0.70924	1.37921	11.69124
downside skew eom	Option	217	10	0.04608	0.79912	0.78729	0.37438	12.75115
12-month mean of realized vol	HAR	217	8	0.03687	0.41229	0.39626	0.57103	18.42396

Notes: Variables are sorted by top-five frequency and then by mean rank. Importance is impurity based and should be interpreted as a relative diagnostic rather than as a structural coefficient.

9.3. Appendix C. Stacked Forecast Diagnostics

The stacked forecasts use the individual forecast panel as inputs to a second-stage combination model. These diagnostics help show which individual forecasts the stacking procedures rely on. The candidate set consists of the 16 individual forecasts used in the current main-table stacked forecasts.

Table C1. Stacked Forecast Candidate Set

Method	Information Set
Elastic Net	HAR
Elastic Net	HAR+Macro
Elastic Net	HAR+Macro+Option
Elastic Net	HAR+Option
Neural Network	HAR
Neural Network	HAR+Macro
Neural Network	HAR+Macro+Option
Neural Network	HAR+Option
PCA	HAR
PCA	HAR+Macro
PCA	HAR+Macro+Option
PCA	HAR+Option
Random Forest	HAR
Random Forest	HAR+Macro
Random Forest	HAR+Macro+Option
Random Forest	HAR+Option

Notes: The candidate forecasts are the individual model-by-information-set forecasts used as inputs to the stacked Elastic Net and stacked Random Forest combinations.

Table C2. Stacked Elastic Net Selected Forecasts

Method	Information set	Months selected	Selection rate	Mean coef.	Share positive	Mean |coef.|	Latest coef.
PCA	HAR+Option	165	0.80097	0.08272	0.99394	0.08344	0.05604
Neural Network	HAR+Macro+Option	150	0.72816	0.06541	1.00000	0.06541	0.04018
Random Forest	HAR+Macro	124	0.60194	0.08237	0.92742	0.08301	-0.01167
Elastic Net	HAR+Macro+Option	109	0.52913	-0.08140	0.18349	0.08502	0.00167
Elastic Net	HAR+Macro	107	0.51942	0.06671	0.98131	0.06776	0.00127
Neural Network	HAR+Macro	103	0.50000	-0.00165	0.52427	0.03678	0.01694
Neural Network	HAR	91	0.44175	0.01765	0.81319	0.03346	0.06333
Random Forest	HAR+Option	83	0.40291	0.08111	1.00000	0.08111	0.00360
Elastic Net	HAR+Option	74	0.35922	0.04753	1.00000	0.04753	0.09280
PCA	HAR+Macro+Option	68	0.33010	-0.04290	0.64706	0.06770	0.00642
Neural Network	HAR+Option	66	0.32039	0.00315	0.54545	0.01709	-0.01801
Random Forest	HAR	62	0.30097	0.00877	0.37097	0.05186	0.01731
PCA	HAR+Macro	59	0.28641	0.01461	0.94915	0.01589	0.00004
Elastic Net	HAR	45	0.21845	0.03984	1.00000	0.03984	0.00047
PCA	HAR	23	0.11165	0.00217	0.91304	0.03717	0.06233
Random Forest	HAR+Macro+Option	8	0.03883	-0.01680	0.75000	0.02380	0.00338

Notes: The table reports individual forecasts with non-zero stacked Elastic Net coefficients. Selection rate is the share of stacked-model refits in which the forecast receives a non-zero coefficient.

Table C3. Stacked Random Forest Forecast Importance

Method	Information set	Months used	Top-5 months	Top-5 rate	Mean imp.	Median imp.	Latest imp.	Mean rank
PCA	HAR+Option	190	190	0.92233	2.51940	2.25742	3.79391	1.41579
Random Forest	HAR+Option	175	117	0.56796	1.53883	1.49671	2.76103	4.98286
Neural Network	HAR+Macro+Option	155	115	0.55825	1.40946	1.30330	1.31854	3.74839
Elastic Net	HAR	206	112	0.54369	1.21582	1.02254	4.67357	6.02427
Elastic Net	HAR+Macro	180	108	0.52427	1.17946	1.13176	1.29216	5.07778
Elastic Net	HAR+Option	195	83	0.40291	1.20782	0.64792	3.71098	7.74872
Random Forest	HAR	206	79	0.38350	1.15350	0.99668	4.91584	6.88835
Random Forest	HAR+Macro	180	72	0.34951	1.01108	0.84727	0.65852	7.43889
PCA	HAR	206	50	0.24272	1.14994	1.02984	4.87837	6.89806
Random Forest	HAR+Macro+Option	155	38	0.18447	0.96949	1.07245	0.98340	7.90968
PCA	HAR+Macro+Option	170	29	0.14078	0.88831	0.87448	0.75798	8.26471
Elastic Net	HAR+Macro+Option	175	21	0.10194	0.60349	0.35013	1.75263	10.46286
Neural Network	HAR	206	10	0.04854	0.78570	0.63797	3.29861	9.57282
Neural Network	HAR+Macro	180	0	0.00000	0.39238	0.43785	0.43927	13.05000
Neural Network	HAR+Option	175	0	0.00000	0.40075	0.31375	1.17197	14.20571
PCA	HAR+Macro	180	0	0.00000	0.27040	0.30484	0.34544	14.71667

Notes: The table reports second-stage Random Forest importance for the individual forecasts. Top-five rate is the share of months in which a forecast ranks among the five most important inputs.